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A Fast and Compact 3-D CNN for Hyperspectral Image Classification

Muhammad Ahmad, Adil Mehmood Khan, Manuel Mazzara, Salvatore Distefano, Mohsin Ali, and Muhammad Shahzad Sarfraz

***Abstract*— Hyperspectral images (HSIs) are used in a large number of real-world applications. HSI classification (HSIC) is a challenging task due to high interclass similarity, high intraclass variability, overlapping, and nested regions. The 2-D convolutional neural network (CNN) is a viable classification approach since HSIC depends on both spectral–spatial information. The 3-D CNN is a good alternative for improving the accuracy of HSIC, but it can be computationally intensive due to the volume and spectral dimensions of HSI. Furthermore, these models may fail to extract quality feature maps and underperform over the regions having similar textures. This work proposes a 3-D CNN model that utilizes both spatial–spectral feature maps to improve the performance of HSIC. For this purpose, the HSI cube is first divided into small overlapping 3-D patches, which are processed to generate 3-D feature maps using a 3-D kernel function over multiple contiguous bands of the spectral information in a computationally efficient way. In brief, our end-to-end trained model requires fewer parameters to significantly reduce the convergence time while providing better accuracy than existing models. The results are further compared with several state-of-the-art 2-D/3-D CNN models, demonstrating remarkable performance both in terms of accuracy and computational time.**

***Index Terms*— 3-D convolutional neural network (CNN), classification, hyperspectral images (HSIs), kernel function.**

I. Introduction

HYPERSPECTRAL sensors collect information (*reflectance*) in several hundreds of contiguous bands with a very high spectral resolution, enabling us to classify objects based on their spectral signatures. However, these images are in relatively low spatial resolution due to the sensor limitations, SNR, and complexity constraints which significantly affect the performance for several real-world applications [1]. The traditional classifiers, for instance, K-nearest neighbour (KNN) [2], support vector machine (SVM) [3], Maximum Likelihood [4], Logistic Regression [2], and extreme learning machine (ELM) [5] only employ spectral information. These classifiers do not perform well due to spectral redundancy and high correlation among the spectral bands. Furthermore, these classifiers fail to preserve the important spatial variability of hyperspectral data which also results in their low performance.

The simplest way to improve the classification performance is to design a classifier that should incorporate both spectral and spatial information. Spatial information is considered as additional discriminatory information associated with the size, shape, and structure of the object which, if provided correctly, brings more competitive results. Spatial–spectral classifiers can generally be classified into two groups. The first category explores spatial and spectral information separately. The spatial information is extracted in advance using entropy [6], morphological operations [7], [8], low rank representation [9], attribute profiles [10], and fuzziness [11]. Later this information is combined with spectral information to perform pixel-level classification.

The second category fuses spatial–spectral information to get joint features [12], for instance, 3-D wavelet, scattering wavelet, and Gabor filter [13], [14] are generated at different frequencies and scales to extract the joint spatial–spectral features for classification. Hyperspectral image (HSI) are in 3-D cubes thus the former category results in several 3-D features, i.e., spatial–spectra feature cubes comprising key information, thus preserving joint spatial–spectral correlations that enable the extracted features to produce better results. However, the classical feature extraction methods are based on shallow learning and handcrafted features which largely depend on domain knowledge [15]. Accordingly, deep models have been used to automatically learn low-to-high-level features from raw HSI data which have attained incredible success for HSI classification (HSIC).

The last few years witnessed an intensive improvement in convolutional neural network (CNN) for HSIC where the spatial features are tailored by a 2-D CNN model [16]–[18]. These spatial features are usually extracted separately that, to some extent, avoid the reason to jointly exploit the spatial–spectral information for HISC. A hybrid spectral CNN for HSIC has been proposed in [15], in which the authors proposed a 3-D CNN followed by a spatial 2-D CNN model. The 3-D convolutional layers facilitate the spectral–spatial feature representation whereas 2-D convolutional layers are

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used to learn abstract level information. The hybrid model produces better results as compared to the conventional 3-D models but still lacks at extracting the abstract level spatial information. Recently, Paoletti *et al.* [19], [20] proposed two deep pyramidal residual networks for HSI feature extraction and classification. The former work only considered spectral information for HSIC whereas the latter considered both the spectral–spatial capsule network for feature learning and classification. Chen *et al.* [21] proposed a 3-D CNN model for feature extraction and classification. Similarly, Zhong *et al.* [22] proposed a spatial–spectral residual network for HSIC in which the residual blocks used identity mapping to connect 3-D convolutional layers. Mou *et al.* [23] proposed an unsupervised HSIC to further explore the residual CNNs.

Our review of the literature revealed several shortcomings, including but not limited to:

1) Though CNNs have become a promising method for HSIC, their memory requirement and high computational complexity makes it challenging to accelerate their performance. This work investigates their application to HSIC targeting high accuracy but under controlled computational cost, in terms of the time it takes for them to converge. To achieve this, our work progressively modifies a baseline model while preserving its accuracy and reducing its time complexity.
2) Preserving channel relationship information is a challenging problem. CNN models are usually trained on reshaped spectral bands or use single band (gray-scale) information (containing different properties), resulting in failure to extract the "fine structural/spatial information of HSI." Furthermore, the high interclass similarity, intraclass variability, overlapping, and nested regions of HSI data makes classification a challenging problem. To overcome the said issue, the proposed architecture first divides the HSI cube into small overlapping 3-D patches. These patches are processed to generate 3-D feature maps using 3-D kernel function over multiple contiguous bands to preserve the joint spatial and spectral information for the feature learning process which exploits important discriminatory information for HSIC.
3) As a preprocessing step, incremental principle component analysis (iPCA) is employed to reduce the redundancy among the bands to process the few important wavelengths out of the entire HSI cube. Finally, to increase the number of spatial–spectral feature maps, four 3-D convolutional layers are deployed to ensure that the model is able to discriminate the spatial information within different spectral bands without any loss.

In a nutshell, our end-to-end trained model requires fewer parameters, which significantly reduces the time it takes for the model to converge without compromising its accuracy, which is better than the existing models, as evident by our experimental results. We performed a comparative study with several state-of-the-art 2-D/3-D CNN methods proposed in the literature, and the experimental/comparative results confirm the superiority of our approach.

The rest of the letter is structured as follows: Section II presents the proposed methodology. Section III describes the experimental Data sets, Results, and discussion. Finally Section IV concludes the letter with possible future research directions.

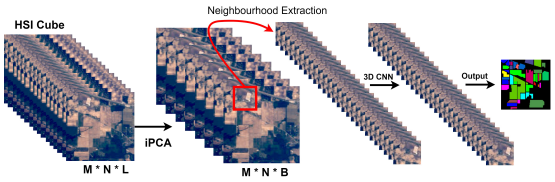


Fig. 1. Proposed 3-D CNN Model for HSIC. 3-D CNN Model details, i.e., the number of 3-D Convectional and fully connected layers, can be found in Table I.

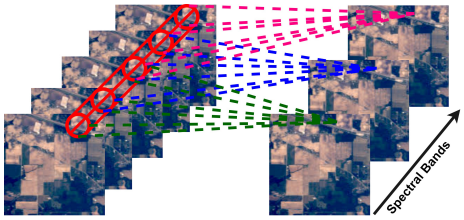


Fig. 2. 3-D convolution operation.

II. Proposed Methodology

Let us assume a HSI can be expressed as $X = [x_1, x_2, x_3, \ldots, x_L]^T \in R^{L\times(N\times M)}$ consisting of $N \times M$ samples associated with C classes per band with total L bands, in which each sample is represented as (x_i, y_j), where y_j is the class label of x_i sample. The HSI pixels exhibit high interclass similarity, high intraclass variability, overlapping, and nested regions. To overcome the aforesaid issues, iPCA is applied to eliminate the redundant bands. iPCA reduces the number of bands (L to B, where $B \ll L$) while maintaining the spatial dimensions as shown in Fig. 1.[1]

In order to pass the HSI cube to the model, it must be divided into small overlapping 3-D spatial patches on which the ground labels are formed based on the central pixel, as shown in Fig. 2. The process creates neighboring patches $P \in R^{S\times S\times B}$ centered at the spatial location (a, b) covering $(S \times S)$ spatial windows [15]. The total of n patches given by $(M - -S + 1) \times (N - -S + 1)$. Thus, these patches cover the width from $(a-(S-1)/2)$ to $(a+(S-1)/2)$ and height from $(b-(S-1)/2)$ to $(b+(S-1)/2)$.

The input patches are first convolved with a 3-D kernel function which computes the sum of the dot product between kernel function and input patch [15], [24]. Later these learned features are processed through an activation function that introduces the nonlinearity. The activation values at spatial position (x, y, z) in the ith layer and jth feature map is denoted as $v_{i,j}^{x,y,z}$. Thus, the final model can be created as follows:

$$v_{i,j}^{x,y,z} = \mathcal{F}\left(\sum_{\tau=1}^{d_{i-1}}\sum_{\lambda=-\nu}^{\nu}\sum_{\rho=-\gamma}^{\gamma}\sum_{\phi=-\delta}^{\delta} w_{i,j,\tau}^{\lambda,\rho,\phi} \times v_{(i-1),\tau}^{(x+\lambda),(y+\rho),(z+\phi)} + b_{i,j}\right)$$

where $\mathcal{F}$ is an activation function, d_{i-1} be the number of 3-D feature maps at $(i-1)$th layer and $w_{i,j}$ be the depth of the kernel, $b_{i,j}$ is the bias, $2\delta+1$, $2\gamma+1$ and $2\nu+1$ be the height, width, and depth of the kernel.

[1] https://github.com/mahmad00/A-Fast-3D-CNN-for-HSIC code is available at GitHub.

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TABLE I

LAYER BASED SUMMARY OF OUR PROPOSED 3-D CNN MODEL ARCHITECTURE SHOWN IN FIG. 2 WITH WINDOW SIZE SET AS 11 × 11

Layer	Output Shape	# of Parameters
Input Layer	(11, 11, 20, 1)	0
Conv3D_1 (Conv3D)	(9, 9, 14, 8)	512
Conv3D_2 (Conv3D)	(7, 7, 10, 16)	5776
Conv3D_3 (Conv3D)	(5, 5, 8, 32)	13856
Conv3D_4 (Conv3D)	(3, 3, 6, 64)	55360
Flatten_1 (Flatten)	(3456)	0
Dense_1 (Dense)	(256)	884992
Dropout_1 (Dropout)	(256)	0
Dense_2 (Dense)	(128)	32896
Dropout_2 (Dropout)	(128)	0
Dense_3 (Dense)	(# of Classes)	774
In total, **994,166** trainable parameters are required		

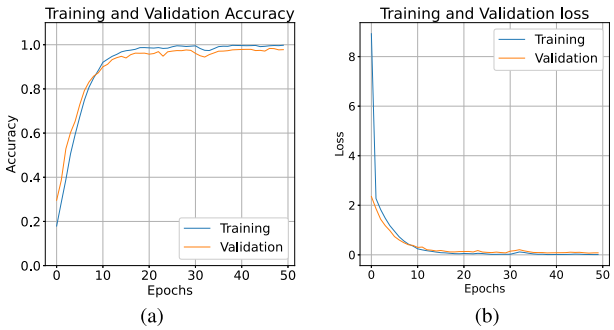
In short, the proposed 3-D CNN convolutional kernels are as follows: $3D_conv_layer1 = 8\times3\times3\times7\times1$ where $K_1^1 = 3, K_2^1 = 3$ and $K_3^1 = 7$. $3D_conv_layer2 = 16 \times 3 \times 3\times5\times8$ where $K_1^2 = 3, K_2^2 = 3$ and $K_3^2 = 5$. $3D_conv_layer3 = 32\times3\times3\times3\times16$ where $K_1^3 = 3, K_2^3 = 3$ and $K_3^3 = 3$ and finally $3D_conv_layer4 = 64\times3\times3\times3\times16$ where $K_1^3 = 3, K_2^3 = 3$ and $K_3^3 = 3$. To increase the number of spatial–spectral feature maps, four 3-D convolutional layers are deployed before the flatten layer to make sure the model is able to discriminate the spatial information within different spectral bands without any loss. Further details regarding the proposed model can be found in Table I. The total number of parameters (i.e., tune-able weights) of our proposed 3-D CNN model is 994 and 166. The weights are initially randomized and optimized using Adam optimizer back-propagation with a soft-max loss function. The weights are updated using a mini-batch of size 256 with 50 epochs without batch normalization and augmentation.

III. EXPERIMENTAL DATA SETS AND RESULTS

The Salinas data set (SD) was acquired over Salinas Valley California using an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. SD is of size 512 × 217 × 224 with a 3.7 m spatial resolution with 512 × 217 spatial and 224 spectral dimensions. SD consists of vineyard fields, vegetables, and bare soils. SD consists of 16 classes. A few water absorption bands 108–112, 154–167, and 224 are removed before analysis.

Indian Pines data set (IPD) is obtained over northwestern Indiana's test site by AVIRIS sensor. IPD is of size 145 × 145 × 224 in the wavelength range 0.4–2.5 × 10^{-6} m where 145 × 145 is the spatial and 224 is the spectral dimensions. IPD consists of 1/3 forest and 2/3 agriculture area and other naturally evergreen vegetation. Some crops in the early stages of their growth are also present with approximately less than 5% of total coverage. Low-density housing, building, and small roads, two dual-lane highway, and a railway line are also a part of IPD. The IPD ground truth (GT) comprises 16 classes which are not mutually exclusive. The water absorption bands have been removed before the experiments, thus the remaining 200 bands are used in this experiment.

Pavia University Data set (PUD) gathered over Pavia in northern Italy using a reflective optics system imaging spectrometer (ROSIS) optical sensor. PUD consists of 610 × 610 spatial and 103 spectral bands with a spatial resolution of 1.3 m. PUD GT classes are 9. Further details about the experimental data sets can be found at [25].



(a) (b)

Fig. 3. Accuracy and loss for training and validation sets on IPD with 11 × 11 window patch correspond to the 50 number of Epochs. (a) Accuracy. (b) Loss.

All the experiments were performed on an online platform known as Google Colab [26]. Google Colab is an online platform that requires a good speed of the internet to run any environment. Google Colab provides an option to execute the codes on python 3 notebook with graphical processing unit (GPU), 25 GB of random access memory (RAM), and 358.27 GB of cold storage for data computation. In all the experiments, the initial Test/Train set is divided into a 30%/70% ratio on which Training samples (70% of the entire population) are further divided into 50%/50% for the Training and Validation set.

For fair comparisons, the learning rate for all the experiments is set to 0.001, $relu$ as an activation function is used for all layers except last on which $softmax$ is used, the patch sizes are set as 11 × 11 × 20, 13 × 13 × 20, 15 × 15 × 20, 17 × 17 × 20, 19 × 19 × 20, 21 × 21 × 20, and 25 × 25 × 20, respectively, with 20 most informative bands selected by the iPCA method. For evaluation purposes, the average accuracy (AA), overall accuracy (OA), and Kappa (κ) coefficient have been computed from the confusion matrices. AA represents the average class-wise classification performance, OA is computed as the number of correctly classified examples out of the total test examples, and finally, κ is known as a statistical metric that considered the mutual information regarding a strong agreement among classification and GT maps. Along with OA, AA, and κ metrics, several statistical tests are also being considered such as F1-Score, Precision, and Recall.

The convergence loss and accuracy of our proposed 3-D CNN model for a 50 number of epochs are shown in Fig. 3. From these figures, one can conclude that the proposed model is converged almost around 20 epochs.

Whereas the computational time of our proposed model is shown in Table II which reveals a fast convergence and computational efficiency of our proposed model. The computational time highly depends on the speed of the internet and available RAM.

The accuracy analysis, i.e., OA, AA, and κ based on the impact of spatial dimensions processed by the proposed model is presented in Table III. While looking into Table III, one can conclude that the window size of 11 × 11 is enough for Pavia University, Salinas, and Salinas-A (SA) data set whereas the window size of 13 × 13 and 25 × 25 both works

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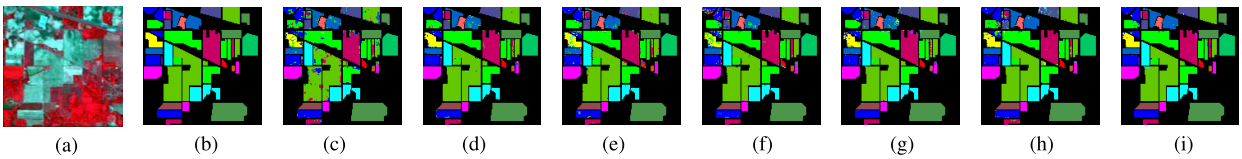


Fig. 4. IPD GTs for each spatial dimensions processed through our proposed model. (a) GT. (b) GT. (c) 11 × 11. (d) 13 × 13. (e) 15 × 15. (f) 17 × 17. (g) 19 × 19. (h) 21 × 21. (i) 25 × 25.

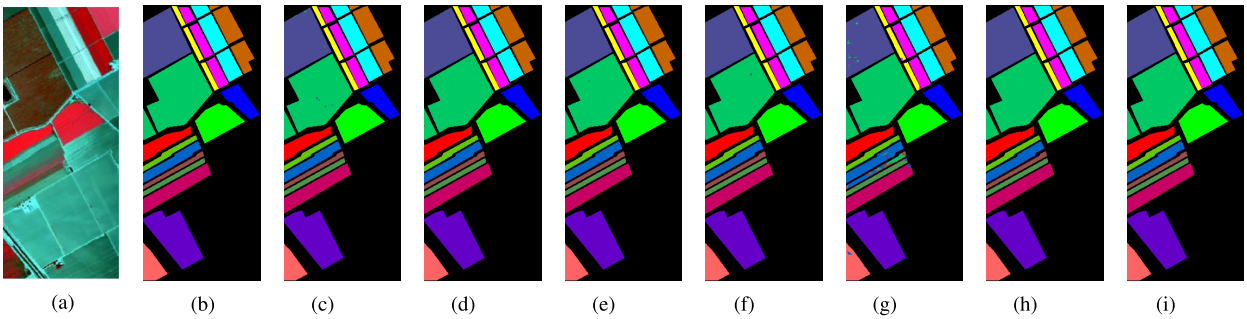


Fig. 5. SD GTs for each spatial dimensions processed through our proposed model. (a) SA. (b) GT. (c) 11 × 11. (d) 13 × 13. (e) 15 × 15. (f) 17 × 17. (g) 19 × 19. (h) 21 × 21. (i) 25 × 25.

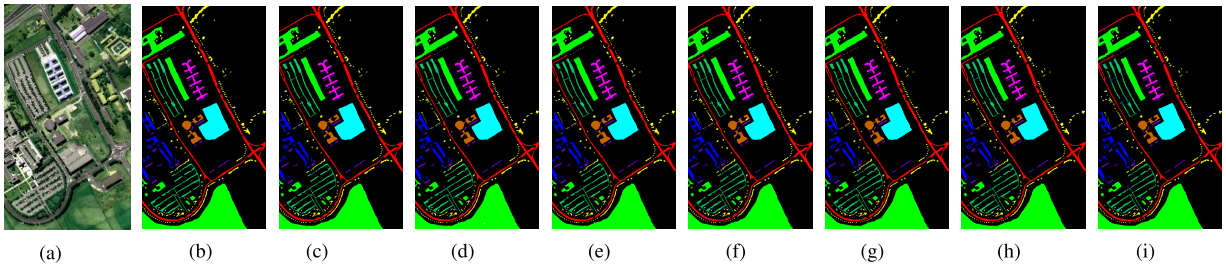


Fig. 6. PUD GTs for each spatial dimensions processed through our proposed model. (a) GT. (b) GT. (c) 11 × 11. (d) 13 × 13. (e) 15 × 15. (f) 17 × 17. (g) 19 × 19. (h) 21 × 21. (i) 25 × 25.

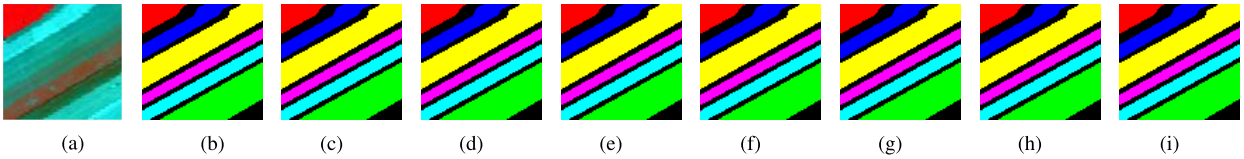


Fig. 7. SA data set GTs for each spatial dimensions processed through our proposed model. (a) GT. (b) GT. (c) 11 × 11. (d) 13 × 13. (e) 15 × 15. (f) 17 × 17. (g) 19 × 19. (h) 21 × 21. (i) 25 × 25.

TABLE II

COMPUTATIONAL TIME IN MINUTES FOR ALL THE EXPERIMENTAL DATA SETS WITH SEVERAL WINDOW SIZES

Dataset	Proposed with Several Window Sizes							2D CNN	3D-CNN	MS-3D-CNN
	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21	25 × 25			
SL-A	0.22	0.23	0.56	0.28	0.98	0.37	0.45	—	—	—
SL	1.34	1.41	1.60	2.00	3.17	2.63	3.52	2.2	74.0	25.5
IP	0.33	0.33	0.61	0.78	0.62	0.58	0.76	1.9	15.2	14.1
PU	2.16	5.26	1.35	2.00	2.46	2.14	2.83	1.8	58.0	20.3

TABLE III

IMPACT OF WINDOW SIZE ON OUR PROPOSED MODEL

Window	PU			IP			SA			SL-A		
	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
11 × 11	99.94	99.89	99.92	88.65	83.52	87.11	99.80	99.91	99.78	100	100	100
13 × 13	99.81	99.65	99.75	95.38	94.14	94.72	99.93	99.94	99.93	100	100	100
15 × 15	99.85	99.62	99.80	93.69	93.09	92.79	99.99	99.99	99.99	100	100	100
17 × 17	99.05	98.49	98.75	91.80	91.74	90.62	99.95	99.97	99.95	99.93	99.93	99.92
19 × 19	99.93	99.78	99.91	93.13	93.42	92.15	98.04	94.02	97.81	100	100	100
21 × 21	99.78	99.43	99.72	94.34	91.31	93.52	99.99	99.99	99.99	100	100	100
25 × 25	98.79	97.67	98.39	97.75	96.17	97.44	99.96	99.93	99.95	100	100	100

almost the same. Furthermore, the classification maps (geographical locations for each class) according to the different number of window sizes (spatial dimensions) are shown in Figs. 4–7. As regards comparison, the proposed model is compared with several state-of-the-art methods published in recent years. From experimental results listed in Table IV one can conclude that the proposed model has competitive results and to some extent better in regards to the other methods. The comparative methods includes multiscale-3-D-CNN [27], 3-D/2-D-CNN [24], [28]. All the comparative methods are being trained according to the settings mentioned in their respective letters. Experiments listed in Table IV show the proposed method improves the results significantly then the

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TABLE IV

COMPARATIVE EVALUATIONS WITH STATE-OF-THE-ART METHODS WHILE CONSIDERING 11×11 SPATIAL DIMENSIONS WITH EVEN LESS NUMBER OF TRAINING SAMPLES (I.E., 60%/40% (TRAIN/TEST) AND 70%/30% (TRAIN/VALIDATION))

Dataset	MS-3D-CNN			3D-CNN			2D-CNN			**Proposed**		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
PU	95.95	97.52	93.40	96.34	97.03	94.90	96.63	94.84	95.53	**98.40**	**97.89**	**97.89**
IP	81.39	75.22	81.20	82.62	76.51	79.25	80.27	68.32	75.26	**97.75**	**94.54**	**97.44**
SA	94.20	96.66	93.61	85.00	89.63	83.20	96.34	94.36	95.93	**98.06**	**98.80**	**97.85**

state-of-the-art methods with even fewer training samples, number of convolutional layers, number of filters, number of epochs, and above all, in less computational time.

IV. CONCLUSION

HSIC is a challenging task due to high interclass similarity, high intraclass variability, overlapping, and nested regions. Though 2-D CNNs have emerged as a viable approach for HSIC, 3-D CNNs are a better alternative because accurate HSIC depends on both spectral–spatial information. However, 3-D CNNs can be highly computational complex due to their volume and spectral dimensions. Therefore, this letter proposed a compact and fast 3-D CNN model, which overcame the above-mentioned challenges. Our model provided state-of-the-art experimental results in a computationally efficient fashion on four hyperspectral benchmark data sets. We resolved the problem of interclass similarity and high intraclass variability using 3-D convolution-based spatial–spectral information. To summarize, our end-to-end trained 3-D CNN has fewer parameters, better recognition accuracy, and fast convergence time than existing 2-D/3-D CNN models. The experimental results reveal that the proposed method outperformed the state of the art methods on various public benchmarks while being less complex than the conventional 3-D CNN models.

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